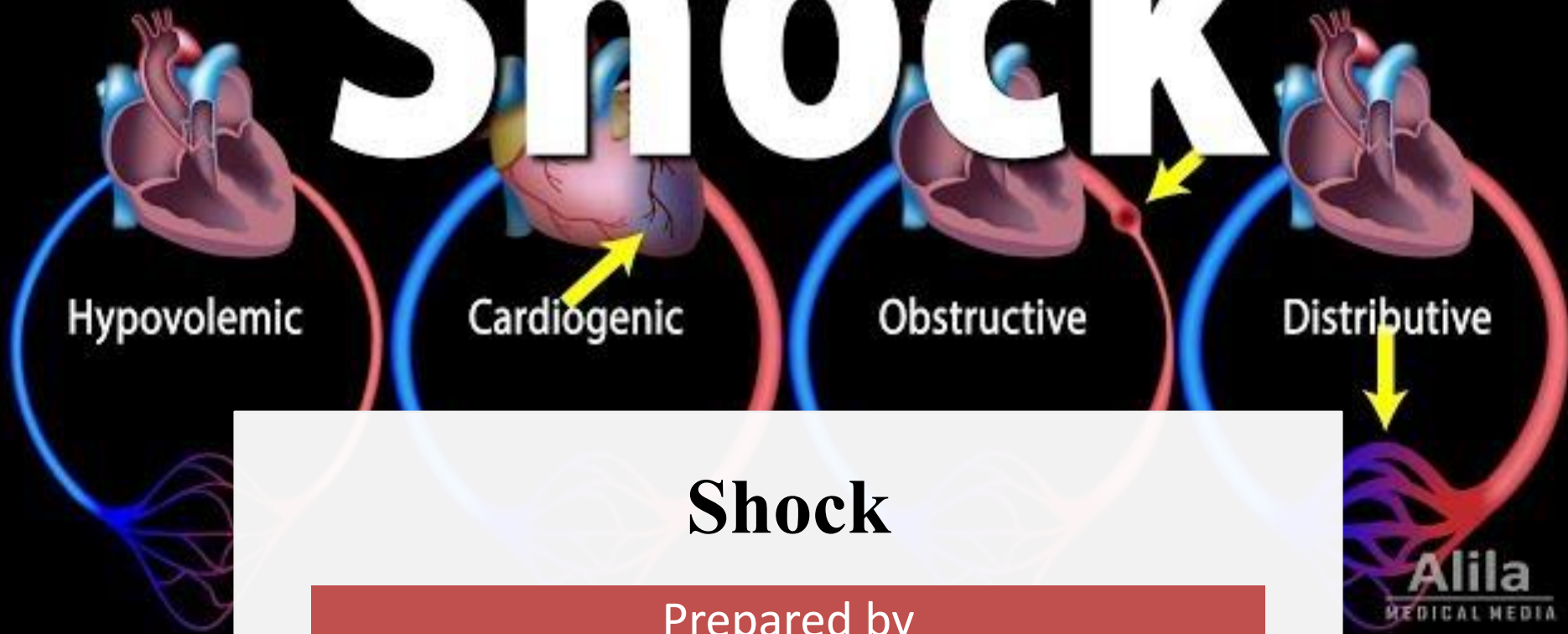


Shock



Shock Syndrome

Outlines:

- Definition of Shock
- Stages of Shock Syndrome
- Types of Shock
- Pathophysiology of Shock
- Assessment and Diagnosis of Shock
- Medical Management of different types of Shock
- Hypovolemic /cardiogenic/septic/anaphylactic /neurogenic Shock
 - Definition
 - Pathophysiology
 - Clinical Manifestation
 - Assessment of hemodynamic parameters
 - Medical and nursing management of different types of shock

Shock

Definition: Shock is an acute, widespread process of impaired tissue perfusion that results in cellular, metabolic, and hemodynamic alterations.

Types of shock

All types of shock eventually result in ineffective tissue perfusion and develop due to imbalance between cellular oxygen supply and cellular demand results in acute circulatory failure or shock syndrome.

Types of Shock

- 1- **Hypovolemic Shock** results from a loss of circulating or intravascular volume.
- 2- **Cardiogenic Shock** results from the impaired ability of the heart to pump.
- 3- **Distributive Shock** results from maldistribution of circulating blood volume and can be further classified as septic, anaphylactic, or neurogenic.

Cont, types of shock

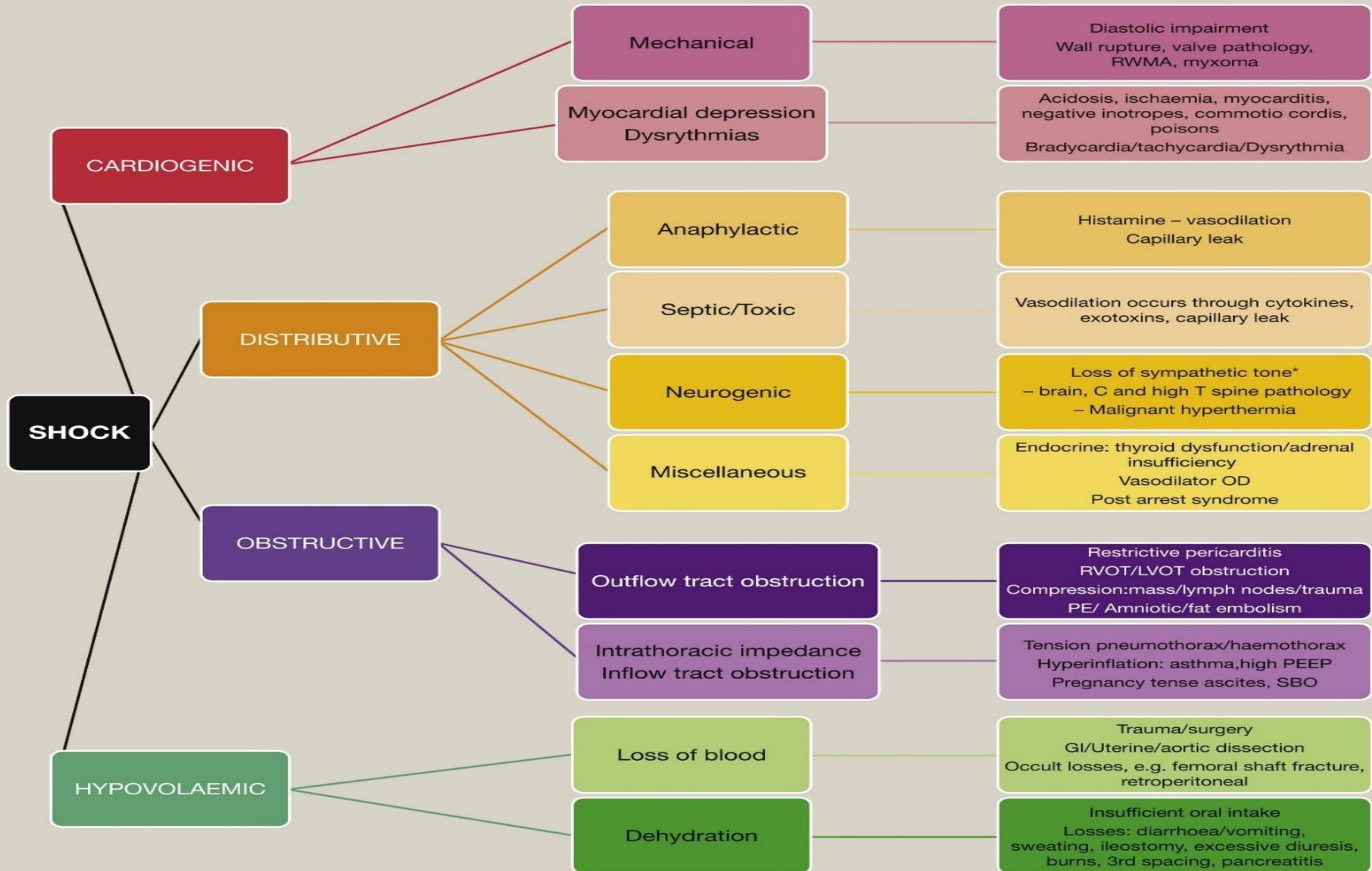
4- **Septic Shock** is the result of microorganisms entering the body.

5-**Anaphylactic** shock is the result of a severe antibody-antigen reaction.

6-**Neurogenic Shock** is the result of the loss of sympathetic tone

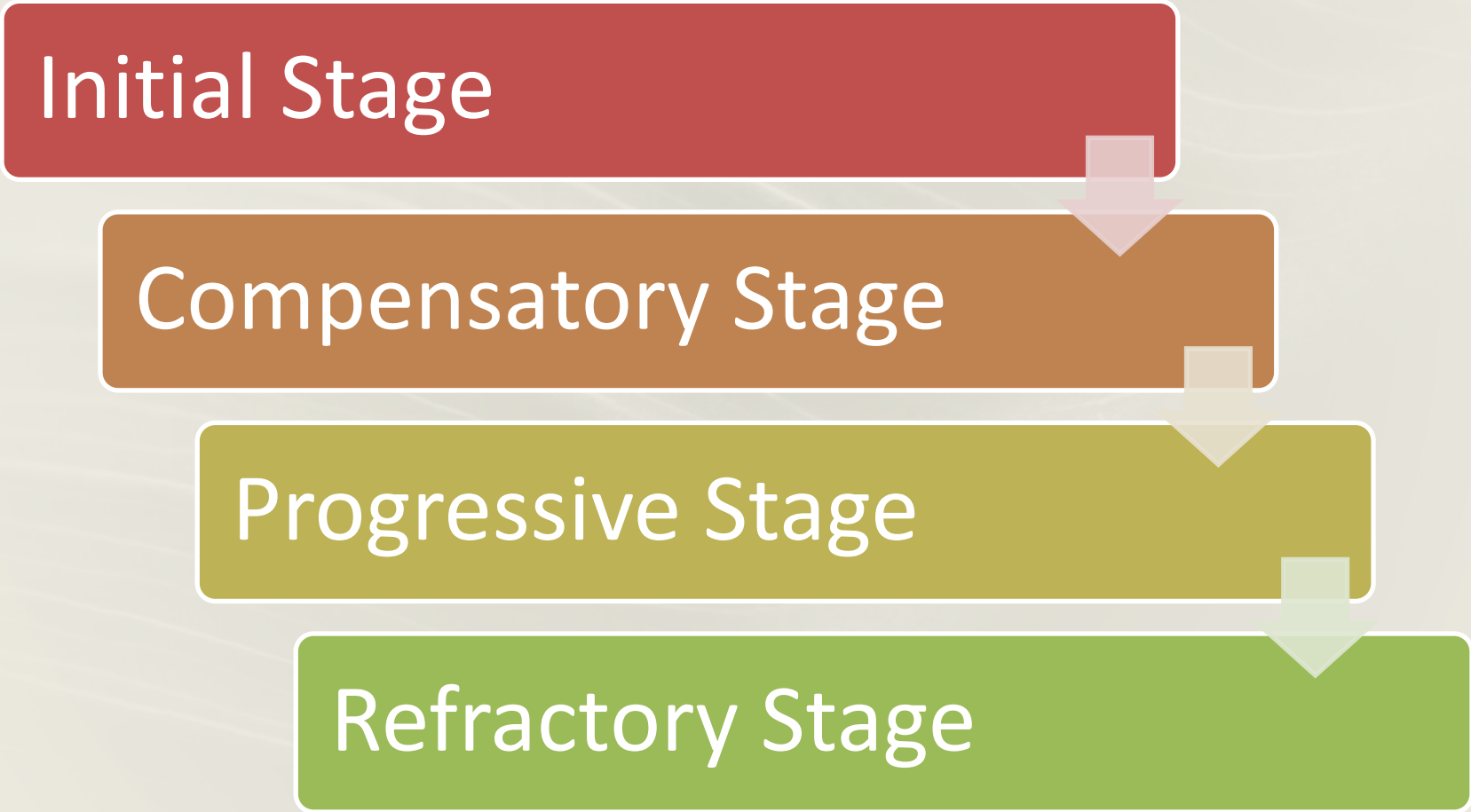
Classification of shock

The clinical classification of different types of shock



Stages of shock

Initial Stage



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graph TD; A[Initial Stage] --> B[Compensatory Stage]; B --> C[Progressive Stage]; C --> D[Refractory Stage];
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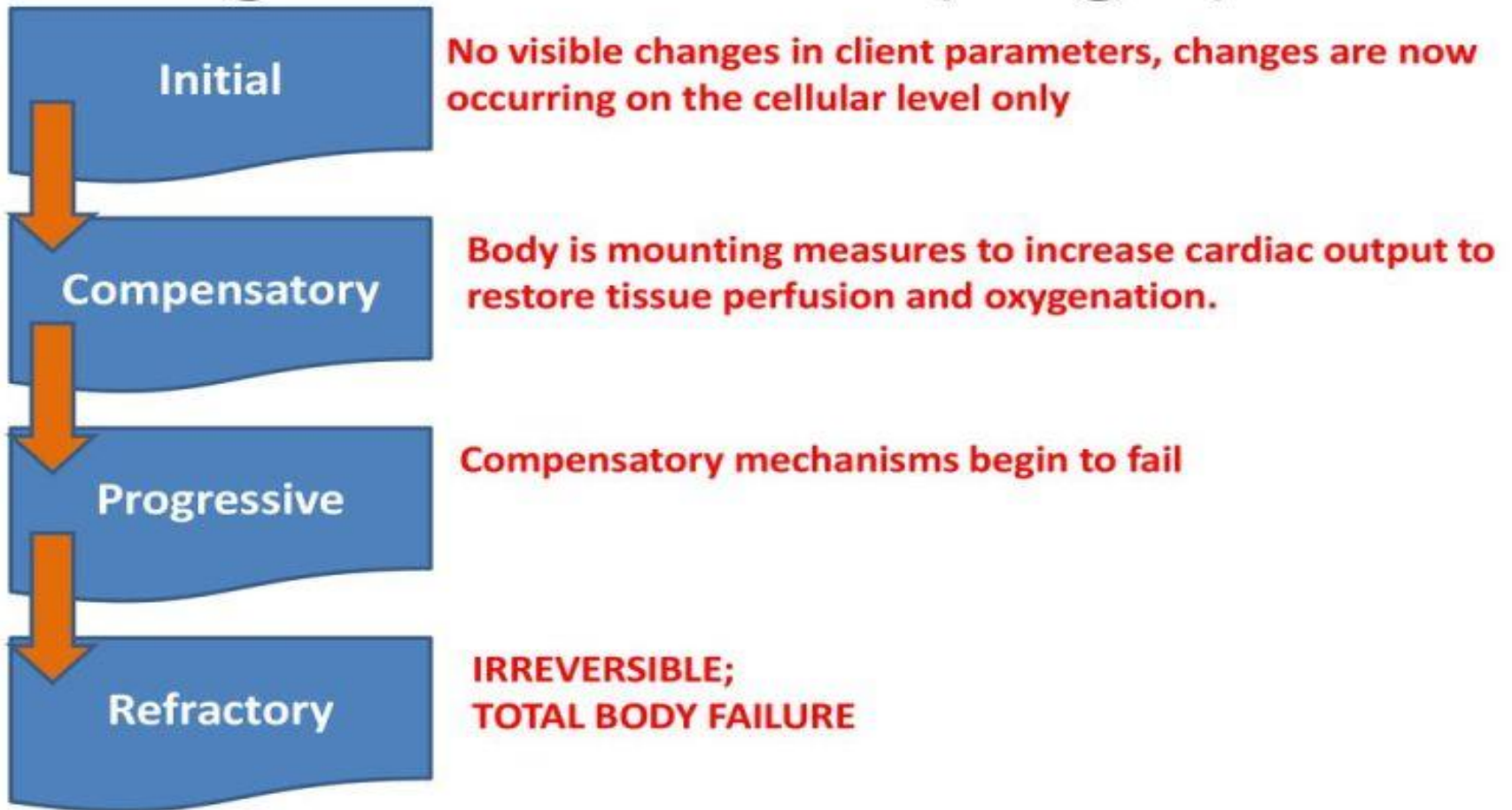
Compensatory Stage

Progressive Stage

Refractory Stage

Stages of shock

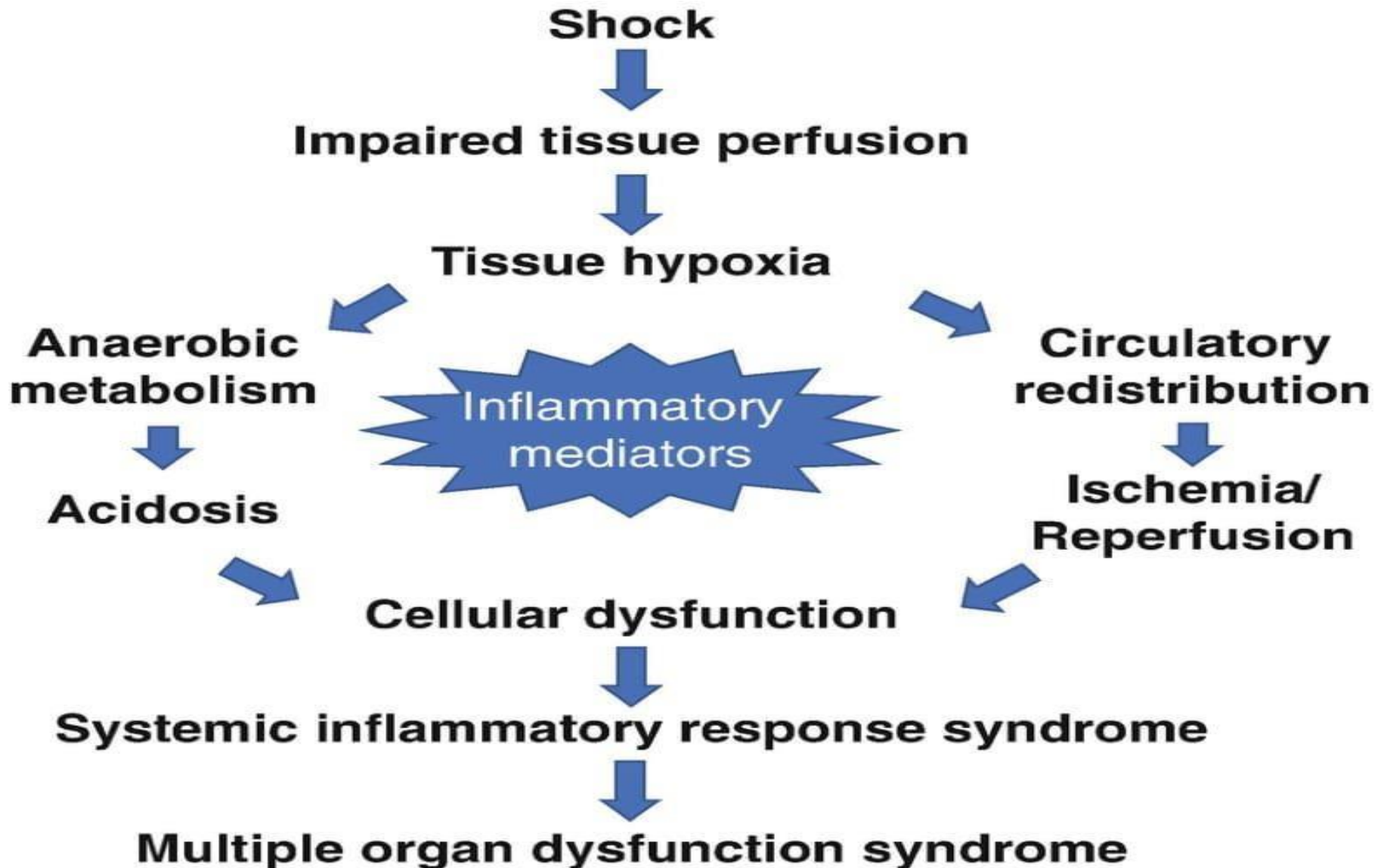
Progression of Shock (Stages)



Pathophysiology of the Initial Stage

- Decreased cardiac output and impaired tissue perfusion causes the cells to switch from Aerobic to Anaerobic metabolism as a source of energy.
- Anaerobic metabolism produces small amounts of energy but large amounts of lactic acid.
- Lactic acidemia causes cellular damage

pathophysiology

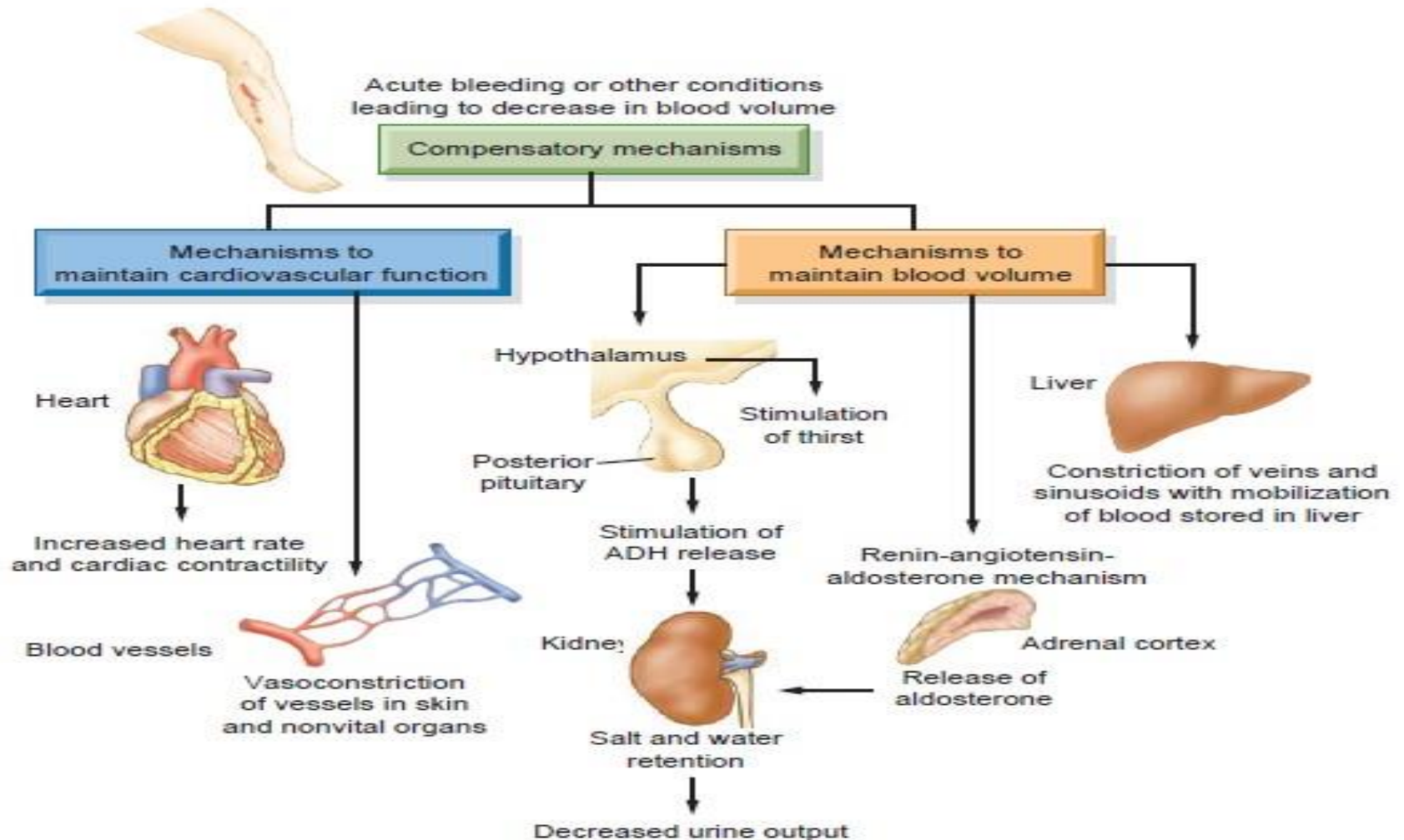


Pathophysiology of the Compensatory Stage

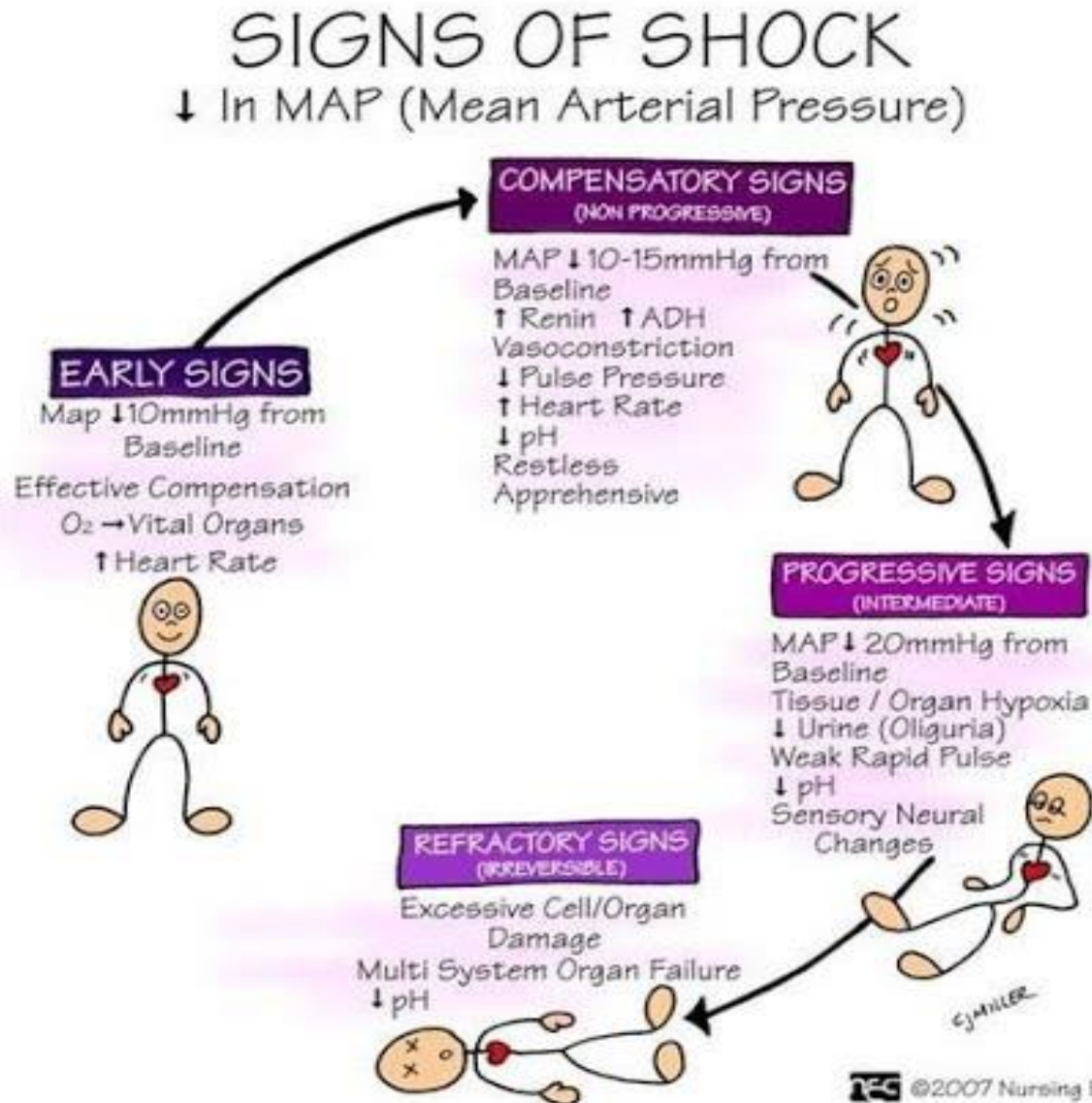
The compensatory mechanisms are mediated by the sympathetic nervous system (SNS) and consist of :

- Neural response
- Hormonal response & Chemical response

Compensatory mechanism of shock



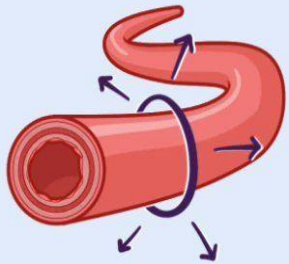
Signs and symptom of shock



Types of shock

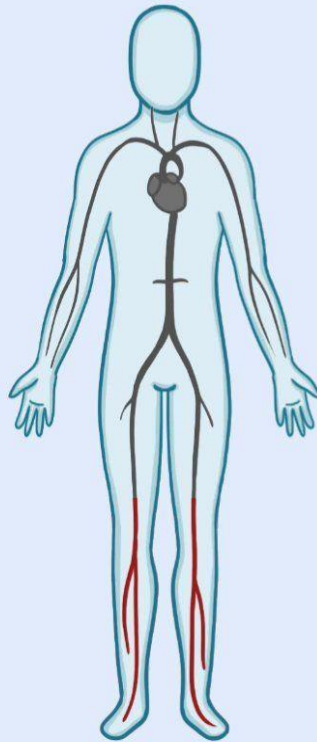
SHOCK **LIFE-THREATENING CONDITION**

DISTRIBUTIVE
EXCESSIVE
VASODILATION

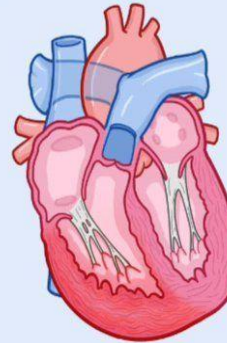


↓
IMPAIRED
BLOOD FLOW
DISTRIBUTION

HYPOVOLEMIC



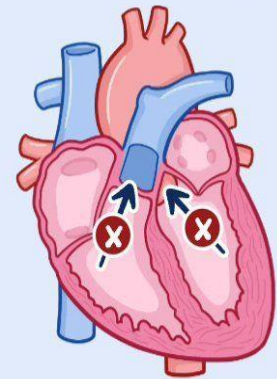
CARDIOGENIC
COMPROMISE of
MYOCARDIAL
PERFORMANCE



↓
↓↓ OUTPUT

OBSTRUCTIVE

OBSTRUCTION
- FILLING HEART
- EJECTING into
GREAT VESSELS

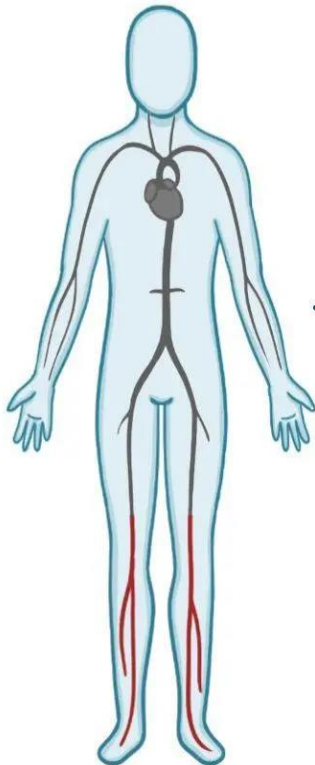


↓
↓↓ OUTPUT

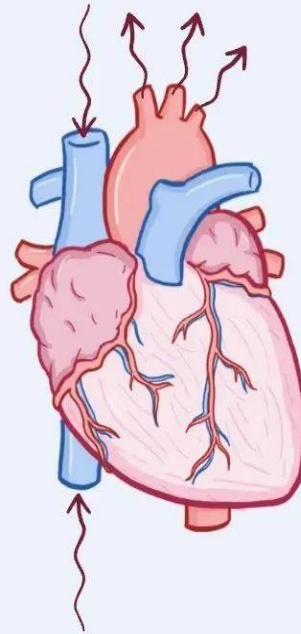
Hypovolemic shock

HYPOVOLEMIC SHOCK **LIFE-THREATENING CONDITION**

↓↓ INTRAVASCULAR
VOLUME



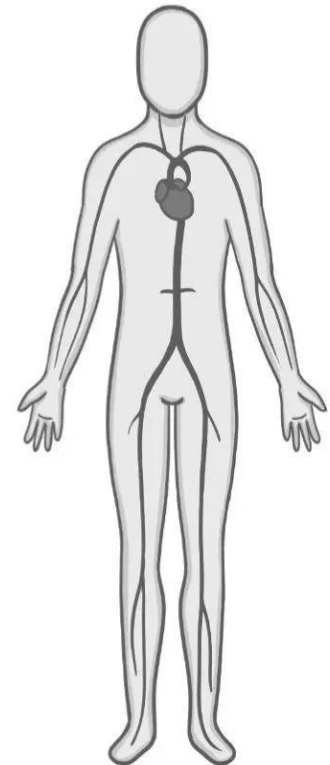
↓↓ VENOUS RETURN



↓↓ CARDIAC OUTPUT

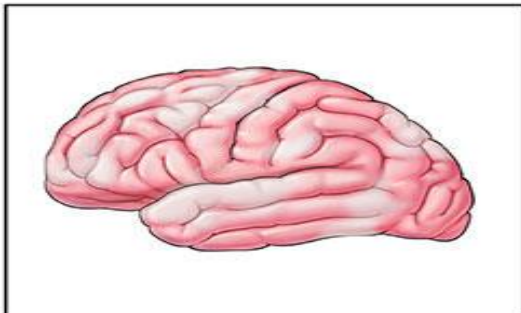
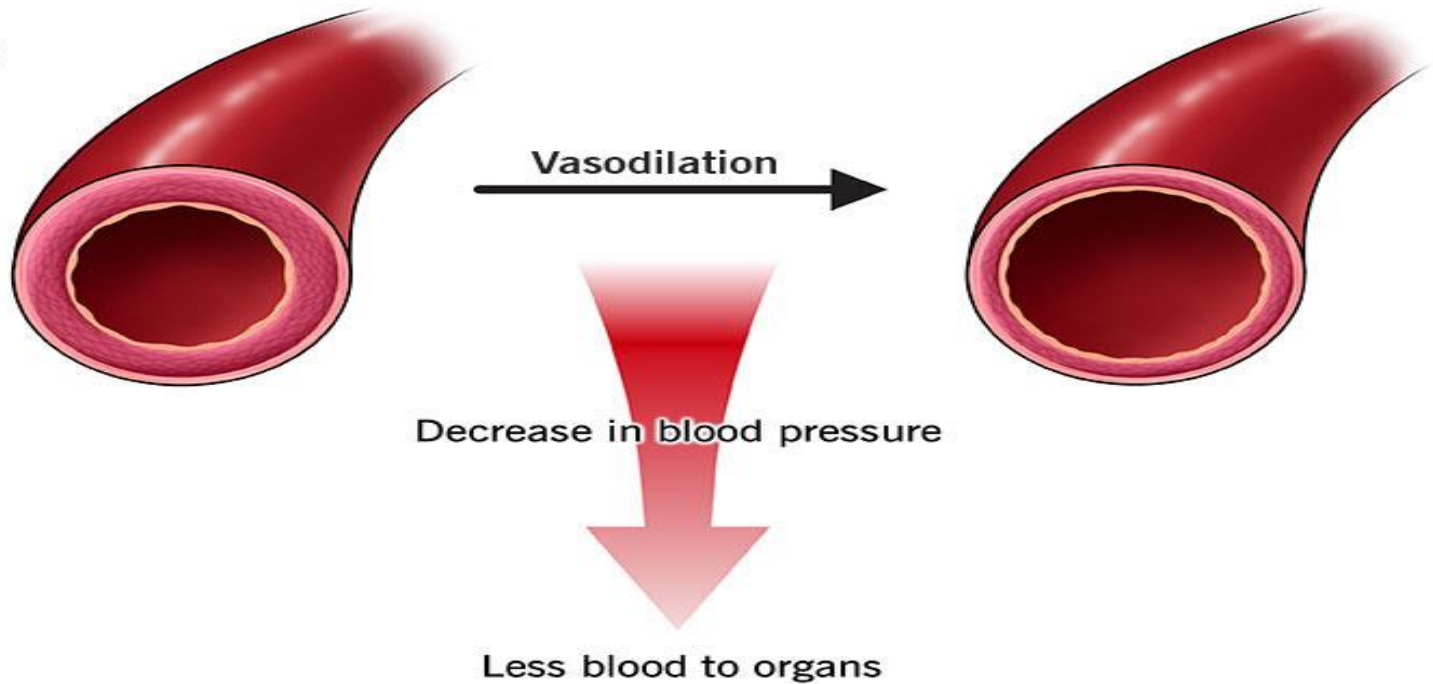


SHOCK

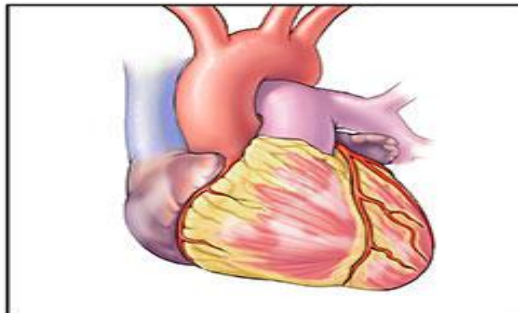


Distributive shock

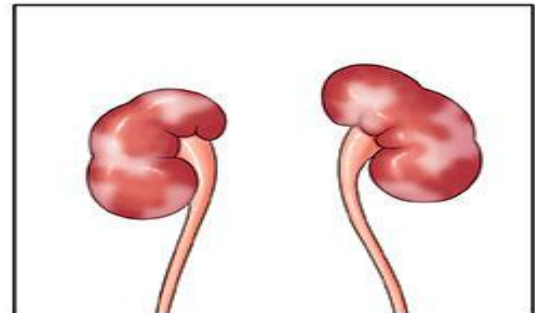
Cleveland
Clinic
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Brain



Heart



Kidneys

Assessment and Diagnosis of shock

- Change in vital signs and hemodynamic parameter due to decrease tissue perfusion and initiation of compensatory mechanism.
- Mean Arterial Blood Pressure (MAP) less than 60 mm Hg and Systolic blood pressure less than 90 mm Hg and tachycardia or bradycardia.

Cont, Assessment

- **Altered mental status** is considered to be a shock state, patient become restlessness, agitation, and then confused and lethargic due to decrease perfusion to the brain.
- **Tachypnea, hypoxemia and worsening chest X-Ray** related to development of acute respiratory distress syndrome due to hypoperfusion.
- **Decrease urine output, rising BUN and creatinine levels** due to initiation of compensatory mechanism and decrease renal perfusion

Cont, Assessment



Petechiae, ecchymosis, bleeding from puncture sites and occult blood in urine, stool, gastric aspirate and tracheal aspirate related to development of disseminated intravascular coagulation.



Hypoglycemia and increase in liver enzymes related to hepatic dysfunction due to hypoperfusion.



Decrease skin turgor and dehydration due to hypovolemia.

Complication of shock

Cardiovascular

- Ventricular failure
- Micro vascular thrombosis

Neurologic

- Sympathetic nervous system dysfunction «
Cardiac and respiratory depression ·
Thermoregulatory failure
- Coma

Pulmonary

- Acute lung failure
- Acute Respiratory Distress Syndrome (ARDS)
Renal

Acute kidney injury

Hematologic

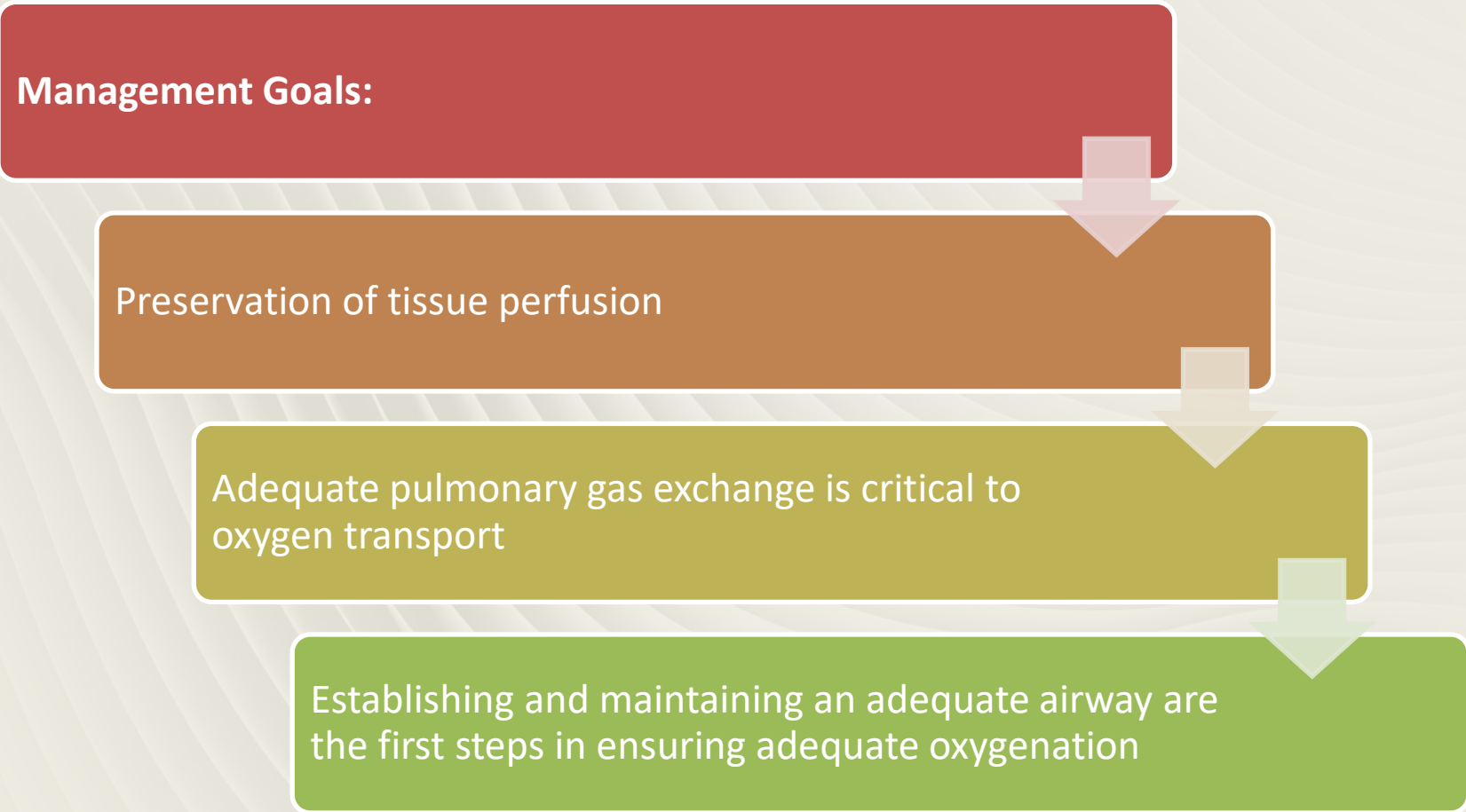
- Disseminated Intravascular Coagulation

Gastrointestinal

- Gastrointestinal tract failure
- Liver failure
- Pancreatic failure

Management of Shock

Management Goals:



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graph TD; A[Management Goals:] --> B[Preservation of tissue perfusion]; B --> C[Adequate pulmonary gas exchange is critical to oxygen transport]; C --> D[Establishing and maintaining an adequate airway are the first steps in ensuring adequate oxygenation];
```

Preservation of tissue perfusion

Adequate pulmonary gas exchange is critical to oxygen transport

Establishing and maintaining an adequate airway are the first steps in ensuring adequate oxygenation

Management Of Shock

Adequate pulmonary gas exchange

1. Establishing and maintaining an adequate airway.
2. After the airway is patent, emphasis is placed on improving ventilation and oxygenation by administration of supplemental oxygen and mechanical ventilator/ support.

Cont, Managment

3- An adequate Cardiac Output (CO) and hemoglobin level are crucial to oxygen transport. CO depends on heart rate, preload, after-load, and contractility.

Cont, management of shock

4- Fluid administration can be accomplished with use of a crystalloid solution, a colloid solution, blood products, or a combination of fluids to restore intravascular volume, maintain oxygen carrying capacity and establish the hemodynamic stability.

5-The choice of fluid and volume and rate of infusion depend on the type of fluid loss, the patient's hemodynamic status and coexisted condition.

Cont, Managment

6-The medications used include vasoconstrictors, vasodilators, positive inotropes, and anti dysrhythmias

Medications

Medication	Action and Use
Dobutamine (Dobutrex)	Stimulates primarily beta1 receptors to increase contractility and increase HR and causes vasodilation in low-CO states
Dopamine (Intropin)	Used in low- CO states or vasodilatory states (distributive shock) to restore vascular tone.
Epinephrine	- Used in hypertensive states to produce constriction of vascular smooth muscle - Promote bronchodilation in anaphylactic shock
Norepinephrine (Levophed)	- Stimulate alpha receptors, which causes vasoconstriction - Used in vasodilatory states (distributive shock) to restore vascular tone. - Stimulation of beta receptors, which increase contractility and HR

Medications for shock

Nitroglycerin

- Vasodilation by direct smooth muscle relaxation, predominantly venous · Used in preload and/ or afterload reduction (cardiogenic shock)
- Dose dependent effect · Arterial dilation only if infusion >1 mcg/kg/min

Nitroprusside (Nipride)

- Vasodilation by direct smooth muscle relaxation, predominantly arterial · Used in preload and/ or afterload reduction (cardiogenic shock)



Evaluate your self

1. What is the main characteristic of shock?

- A. Increased cardiac output
- B. Normal tissue perfusion
- C. Increased urine output
- D. Decreased tissue perfusion**

2. Which type of shock results from severe blood or fluid loss?

- A. Cardiogenic shock
- B. Septic shock
- C. Hypovolemic shock**
- D. Neurogenic shock

Evaluate your self

3. The most common cause of cardiogenic shock is:

- A. Pulmonary embolism
- B. Myocardial infarction**
- C. Spinal cord injury
- D. Anemia

4. The primary problem in all types of shock is:

- A. Decreased tissue perfusion**
- B. Increased blood flow
- C. High blood pressure
- D. Increased oxygen delivery

